BT402 (Microbial Biotechnology)

Final Term (Current Papers 2021)

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We Grow Together

What is DNA vaccine?

A DNA vaccine is a type of vaccine that transfects a specific antigen-coding DNA sequence into the cells of an organism as a mechanism to induce an immune response. These vaccines show great promise and several types are being tested in humans. DNA vaccines take immunization to a new technological level. These vaccines dispense with both the whole organism and its parts and get right down to the essentials: the microbe's genetic material. Example: Influenza vaccine.

What are Conjugate Vaccines?

If a bacterium possesses an outer coating of sugar molecules called polysaccharides, as many harmful bacteria do, researchers may try making a conjugate vaccine for it. Polysaccharide coatings disguise a bacterium's antigens so that the immature immune systems of infants and younger children can't recognize or respond to them. Example : Haemophilus influenza type B vaccine.

What are Live Attenuated Vaccines? Give example.

Live, attenuated vaccines contain a version of the living microbe that has been weakened in the lab so it can't cause disease. Because a live, attenuated vaccine is the closest thing to a natural infection, these vaccines are good "teachers" of the immune system. Example: Vaccines against measles, mumps, and chickenpox.

What are Subunits Vaccines?

Instead of the entire microbe, subunit vaccines include only the antigens that best stimulate the immune system. In some cases, these vaccines use epitopes the very specific parts of the antigen that antibodies or T cells recognize and bind to. Immunization because subunit vaccines contain only the essential antigens and not all the other molecules that make up the microbe. Example: Plague immunization.

Define Adjuvant?

An adjuvant is a substance that is formulated as part of a vaccine to enhance its ability to induce protection against infection. The word "adjuvant" comes from the Latin adjuvare and means "to

help." Adjuvants help activate the immune system, allowing the antigens pathogen components that elicit an immune response in vaccines to induce long-term protective immunity. An adjuvant is an ingredient used in some vaccines that helps create a stronger immune response in people receiving the vaccine. In other words, adjuvants help vaccines work better.

What are the Key role of adjuvant in modern vaccine?

Vaccines containing novel adjuvant formulations are increasingly reaching advanced development and licensing stages, providing new tools to fill previously unmet clinical needs. However, many adjuvants fail during product development owing to factors such as manufacturability, stability, lack of effectiveness, unacceptable levels of tolerability or safety concerns.

What are the challenges with adjuvant?

Lower reactogenicity adjuvants came with reduced vaccine efficacy. Safety and tolerability are critical regulatory issues facing the adjuvants. In most cases increased adjuvant potency is associated with increased reactogenicity and toxicity.

What is Microbial mining or Biomining?

The use of microorganisms to remove minerals particularly metals from rocks. Biomining is a technique of extracting metals from ores and other solid materials typically using prokaryotes or fungi.

Why microorganisms are said to be chemolithotrophic?

The microorganisms are said to be chemolithotrophic ("rock eating") because they obtain energy from the oxidation of inorganic substances.

Define Bioleaching?

Bioleaching is a process in mining that extracts valuable metals from a low-grade ore with the help of microorganisms such as bacteria or archaea. The microorganisms in the river water were breaking down the sulfide-containing ores to extract copper. The microorganisms that played a role in the dissolution of metals from their ores, a process termed "bioleaching".

How biofuel generate charging electronics?

According to scientists from Saint Luis University, a fuel cell was developed with cooking oil and sugar to generate electricity consumers will be able to use these cells instead of generating electricity. Consumers may be able to use fuel cells in place of batteries to charge anything from computers to cell phones.

What is compost?

Compost is organic matter that has been decomposed and recycled as a fertilizer and soil amendment. Compost is a key ingredient in organic farming. End product of the decomposition of organic matter.

Write down the uses of antimicrobials?

- One recent study indicated that personal care products such as soaps, body washes and hand sanitizers that contain antimicrobial chemicals are more effective at limiting bacteria than soaps without antimicrobial agents this 2011 study found that washing hands with an antimicrobial soap reduced bacteria on skin at a greater rate than non-antimicrobial soap.
- Triclosan, an antimicrobial compound, has been found to promote oral health and aid in the fight against gingivitis when added to toothpaste.
- Kitchen counters, office desks, bathroom sinks, and other high-traffic areas in homes and offices may accumulate germs that can make people sick.
- Cleaning products without antibacterial/antimicrobial ingredients will clean the surface, but will not kill the germs on it.
- Antimicrobial pesticides help to slow the growth of human pathogenic microorganisms and are used in food processing plants, dairies, breweries, poultry houses and other animal feeding operations, hospitals and medical and dental clinics and offices, municipal drinking water and water treatment facilities, swimming pools and spas, schools, day-care centres, public access facilities and homes.

Define Antivirals?

Antiviral drugs are a class of medication used specifically for treating viral infections. Like antibiotics, specific antivirals are used for specific viruses. They are relatively harmless to the host and therefore can be used to treat infections.

What are antimicrobials or antimicrobials pesticides?

Antimicrobial products kill or slow the spread of microorganisms. Microorganisms include bacteria, viruses, protozoans, and fungi such as mold and mildew. They are crucial in helping to prevent and stop the spread of harmful microbes.

What are the benefits of composting?

Soil conditioner

With compost, you are creating rich humus for lawn and garden. This adds nutrients to your plants and helps retain moisture in the soil.

Recycles kitchen and yard waste

Composting can divert as much as 30% of household waste away from the garbage can.

Introduces beneficial organisms to the soil

Microscopic organisms in compost help aerate the soil, break down organic material for plant use and ward off plant disease.

Good for the environment

Composting offers a natural alternative to chemical fertilizers.

Reduces landfill waste

Most landfills in North America are quickly filling up; many have already closed down. One-third of landfill waste is made up of compostable materials.

What is vaccine?

A vaccine is a biological preparation that improves immunity to a particular disease. A vaccine typically contains an agent that resembles a disease-causing microorganism and is often made from weakened or killed forms of the microbe. The agent stimulates the body's immune system to recognize the agent as foreign, destroy it, and keep a record of it. So that the immune system can more easily recognize and destroy any of these microorganisms that it later encounters. The terms vaccine and vaccination are derived from Variolae vaccinae (smallpox of the cow), the term devised by Edward Jenner to denote cowpox.

What are the history behind the production of vaccine?

During the late 1760s whilst serving his apprenticeship as a surgeon Edward Jenner learned of the story, common in rural areas, that dairy workers would never have the often-fatal or disfiguring disease smallpox because they had already had cowpox, which has a very mild effect in humans. In 1796, Jenner took pus from the hand of a milkmaid with cowpox, scratched it into the arm of an 8-year-old boy. Six weeks later inoculated the boy with smallpox, afterwards observing that he did not catch smallpox. Jenner extended his studies and in 1798 reported that his vaccine was safe in children and adults. In 1885 the French microbiologist Louis Pasteur and Emile roux developed the first vaccine against Rabies.

Objectives of Transgenic Plants?

The main objectives in the production of transgenic plants are attempts to produce crops, which have ideal traits, quality, and high yield or engineer metabolic pathways for the production of tailor-made plant polymers or low molecular weight compounds, increased resistance towards pathogens and pesticides, improved food quality, and the production of polypeptides for pharmaceutical or technical use. Plant-made vaccines or antibodies are especially attractive as plants are free of human diseases, reducing screening costs for viruses and bacterial toxins.

How animals digest the cellulose and how they get from?

Cellulose is hard to digest by the animal. Cellulose can be digest by enzyme named cellulase. Herbivores depend on bacteria that live in their stomach to digest cellulose. This bacteria secretes (merembeskan) cellulase into the stomach.

Write the Importance of microbes in agriculture and medicine ?

Agriculture

- Bacteria play important roles to promote the plant growth. Bacteria, Break down dead plants and dead animals by releasing ammonia. Ammonia adds nitrogen into the soil. The nitrogen fixing bacteria in the soil change the ammonia into nitrates. This nitrates will be used by plants.

Medicine

- Some type of microorganism produce important drugs called antibiotics. Functions of antibiotics weaken, destroy other microorganism such as Penicillium notatumis a green mould. Its produces antibiotic called penicillin.
- Microbes play defence
- Microbes boost the immune system
- Microbes protect us from auto-immune diseases.
- Microbes detoxify and may even fight off stress.

Define Antifungals?

Antifungals are used to kill or prevent further growth of fungi. In medicine, they are used as a treatment for infections such as athlete's foot, ringworm and thrush and work by exploiting differences between mammalian and fungal cells. They kill off the fungal organism without dangerous effects on the host.

What are the types of antimicrobials with associated examples.

Antibacterials

Antibacterials are used to treat bacterial infections. The drug toxicity to humans and other animals from antibacterials is generally considered low. Antibacterials are among the most commonly used drugs and among the drugs commonly misused by physicians, for example, in viral respiratory tract infectiously low. Prolonged use of certain antibacterials can decrease the number of gut flora which may have a negative impact on health.

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Antivirals

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Antiparasitics

Antiparasitics are a class of medications indicated for the treatment of infection by parasites, such as nematodes, cestodes, trematodes, infectious protozoa, and amoebae. Like all antimicrobials against intracellular microbes, they must kill the infecting pest without serious damage to the host.

What are Pesticides?

Pesticides are chemical substances that are meant to kill pests. In general, a pesticide is a chemical or a biological agent such as a virus, bacterium, antimicrobial, or disinfectant that deters, incapacitates, kills, pests.

What is Microbial degradation?

Microbial biodegradation is the use of bioremediation and biotransformation methods to harness the naturally occurring ability of microbial xenobiotic metabolism to degrade, transform or accumulate environmental pollutants, including hydrocarbons (e.g. oil), polychlorinated biphenyls (PCBs), polyaromatic hydrocarbons.

Use of bacteria in agricultural field?

Agriculture

Bacteria play important roles to promote the plant growth. Bacteria break down dead plants and dead animals by releasing ammonia. Ammonia adds nitrogen into the soil. The nitrogen fixing bacteria in the soil change the ammonia into nitrates. This nitrates will be used by plants.

Define genomics?

Genomics is the study of the entire set of genes in the genome of a cell.

Enlist the methods of applying Bio-fertilizers on crop and also write them in detail.

Seedling root dip

This method is applied to the rice crop. A bed of water is spread on the land where the crop has to grow. The seedlings of rice are planted in the water and kept there for eight to ten hours.

Seed treatment

Within this process your nitrogen along with phosphorus fertilizers are generally put together jointly throughout normal water. And then vegetables are generally dropped within this mix. Following purposes on this substance on the vegetables, vegetables are generally dried up. As soon as they normally dry out, they must always be sown immediately ahead of that they find harmed by simply unsafe bacteria.

Soil treatment

All the bio fertilizers along with the compost fertilizers are mixed together. They are kept for one night. Then the next day the mixture is spread on the soil where seeds have to be sown.

Types of fermentation

There are two types of fermentation

- Lactic acid fermentation occurs in animal muscles when the tissue requires energy at a faster rate than oxygen can be supplied and used to convert lactose into lactic acid in yogurt and cheese production and also gives the sour taste to fermented vegetables such as pickles.
- Alcoholic fermentation done by Yeast and certain bacteria. Pyruvate is broken into alcohol and carbon dioxide and used in the production of beer, wine and bread.

Why plants need nitrates?

Plants need nitrates to make amino acids. E.g : Pea and bean plants use bacteria Rhizobium which directly convert nitrogen into acid amino.

Write the production of bread?

Yeast causes the bread to rise by producing carbon dioxide from the carbohydrates in the dough. Carbon dioxide gas bubbles released in the dough causes it to rise. So the bread become fluffy.

How protein function by interacting with other proteins?

Proteins bind to each other through a combination of hydrophobic bonding, van der Waals forces, and salt bridges at specific binding domains on each protein.

Difference between bioleaching and microbial mining?

Bioleaching

Bioleaching is a process in mining that extracts valuable metals from a low-grade ore with the help of microorganisms such as bacteria or archaea. The microorganisms in the river water were breaking down the sulfide-containing ores to extract copper. The microorganisms that played a role in the dissolution of metals from their ores, a process termed “bioleaching”.

Microbial mining

The use of microorganisms to remove minerals particularly metals from rocks. Bio-mining is a technique of extracting metals from ores and other solid materials typically using prokaryotes or fungi.

Define prebiotic?

A non- digestible food ingredient that beneficially affects the host by selectively stimulating the growth and/ or activity off one or a limited number of bacteria in the colon, and thus Improves host health.

How prebiotic are produced?

Prebiotics are indigestible oligosaccharides commonly produced from food sources such as chicory root, milk, and oats and are known to promote the growth of beneficial probiotic microorganisms in the lower intestine of mammals.

Write down the two types of prebiotic?

Difference between metagenomics and metabolomics?

Metagenomics

Metagenomics is the study of genetic material recovered directly from environmental samples.

Metabolomics

Metabolomics is the large-scale study of small molecules, commonly known as metabolites, within cells, bio fluids, tissues or organisms.

What is probiotic culture?

Probiotic cultures

Probiotic cultures have become popular in dairy products because of their health benefits. These cultures are all very carefully selected strains, and there is good evidence that they help improve digestion, protects the immune system and keep the body's intestinal flora in balance.

Names of two pharmaceutical drugs and how they are produced?

Many pharmaceutical drugs are also produced by microbes e.g. Cyclosporin A, that is used as an immunosuppressive agent in organ-transplant patients, is produced by the fungus Trichoderma polysporum.

Statins produced by the yeast Monascus purpureus have been commercialised as blood cholesterol-lowering agents. It acts by competitively inhibiting the enzyme responsible for the synthesis of cholesterol.

What is MALDI?

MALDI is the abbreviation for "Matrix Assisted Laser Desorption/Ionization.“ The sample for MALDI is uniformly mixed in a large quantity of matrix. The matrix absorbs the ultraviolet light (nitrogen laser light, wavelength 337 nm) and converts it to heat energy. A small part of the matrix (down to 100 nm from the top outer surface of the Analyte) heats rapidly (in several Nano-seconds) and is vaporized, together with the sample.

What are protein arrays?

Protein microarray is an arrangement of purified proteins on a solid surface like a glass slide.

Define Probiotics?

Probiotics are live bacteria and yeasts promoted as having various health benefits. They're usually added to yoghurts of taken as food supplements, and are often described as good" or " friendly" bacteria.

Define Synbiotics?

Synbiotics are mixtures of probiotics (helpful gut bacteria) and prebiotics (non-digestible fibers that help these bacteria grow).

Name three bacteria which is used in the food production like bread and Yogurt?

- ✓ Yeast
- ✓ Penicillium
- ✓ Lactobacillus

Names of two pharmaceutical drugs?

- Cyclosporin A
- Statins

What is Biopesticides and Bio-fertilizers?

Biopesticides

Bio pesticide is a formulation made from naturally occurring substances that controls pests by non toxic mechanisms and in eco-friendly manner. Biopesticides are certain types of pesticides derived from such natural materials as animals, plants, bacteria, and certain minerals.

Biofertilizers

Biofertilizers are substances that contain living microorganisms, which when added to the soil increase its fertility and promotes plant growth.

Role of microbes in food supply?

Microbes challenge the food supply by causing yield-reducing diseases in food producing plants and animals.

Define Synbiotics? How Synbiotic prevent to diseases?

Synbiotics

Synbiotics are mixtures of probiotics (helpful gut bacteria) and prebiotics (non-digestible fibers that help these bacteria grow).

Prevent to diseases :

Synbiotics are a combination of prebiotics and probiotics that are believed to have a synergistic effect by inhibiting the growth of pathogenic bacteria and enhancing the growth of beneficial organisms.

Mechanism for regulatory processes?

A regulatory mechanism is the combination of steps or processes that an organism can engage to ensure that a biological process is controlled.

Microbes in Meat Industry?

Meat starter cultures are used to make dried, fermented products such as salami, Salami is a type of cured sausage consisting of fermented and air-dried meat, typically beef or pork. Pepperoni, Pepperoni is an American variety of salami, made from cured pork and beef mixed together and seasoned with paprika or other chili pepper. Dried ham, Ham is pork that has been preserved through salting, smoking, or wet curing. Lactic bacteria develop the flavour and colour of the products. In addition, a wide variety of moulds are used to ripen the surface of sausages preserving the natural quality of the product and controlling the development of flavour.

What are Biofertilizers?

“Biofertilizers are substances that contain microorganisms, which when added to the soil increase its fertility and promotes plant growth.”

Importance of Biofertilizers

- Biofertilizers improve the soil texture and yield of plants.
- They do not allow pathogens to flourish.
- They are eco-friendly and cost-effective.
- Biofertilizers protect the environment from pollutants since they are natural fertilizers.

- They destroy many harmful substances present in the soil that can cause plant diseases.
- Biofertilizers are proved to be effective even under semi-arid conditions.

Applications of Biofertilizers

Following are the important applications of biofertilizers:

- **Seedling root dip**

This method is applicable to rice crops. The seedlings are planted in the bed of water for 8-10 hours.

- **Seed Treatment**

The seeds are dipped in a mixture of nitrogen and phosphorus fertilizers. These seeds are then dried and sown as soon as possible.

- **Soil Treatment**

The biofertilizers along with the compost fertilizers are mixed and kept for one night. This mixture is then spread on the soil where the seeds have to be sown.